

AI for Compliance

BISfit

BIS Standards Recommendation Engine
for Indian Micro & Small Enterprises



BIS STANDARDS
MSME RAG System

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01 | Problem Statement

The MSE Compliance Gap

- > 63 million MSEs in India face complex BIS compliance requirements
- > Weeks spent manually identifying applicable standards
- > 500+ BIS standards in Building Materials alone
- > Non-compliance risks: penalties, market rejection, safety failures
- > No affordable, accessible tool exists for smaller enterprises

6–8 Weeks

Average time to identify applicable BIS standards

63M+

MSEs in India needing compliance guidance

₹2L–5L

Estimated cost of non-compliance per incident



BISfit — AI Solutions for Indian Construction Standards

A Production-Ready RAG System based on SP 21 : 2005 (Summaries of Indian Standards for Building Materials)

PROBLEM & SOLUTION



PROBLEM

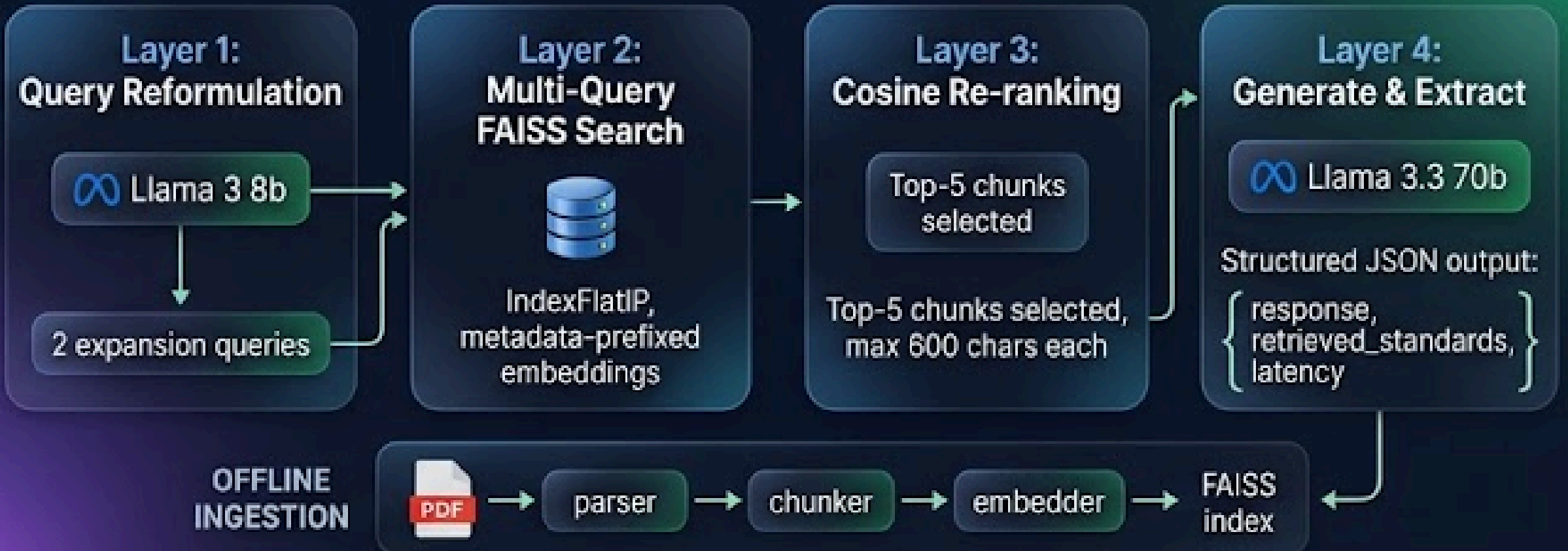
Hours spent manually searching 580+ Indian Standards.



SOLUTION

Natural language AI, narrative answers, exact IS codes, sub-2s average response.

SYSTEM ARCHITECTURE: 4-LAYER INFERENCE PIPELINE



USER PROFILES



Manufacturer
Which standard governs my product?



Site Engineer
Comp. strength for 53 grade OPC?



QA Inspector
Hollow concrete masonry tests?



Architect
Corrugated asbestos sheets?

KEY DESIGN DECISIONS

Domain-aware chunking:
1 IS = 1 Chunk

Metadata-prefixed
Embeddings

Round-robin key rotation

Single combined LLM call

PERFORMANCE METRICS

~90% Hit@3

~1.7s Latency chip ⚡

MRR@5 ~0.90

TECH STACK



Groq LPU



Llama 3
(8b, 70b)



PyMuPDF



FAISS



FastAPI



Vanilla
HTML/CSS/JS

EVALUATION-READINESS

Correctness

Speed

Robustness

Isolation

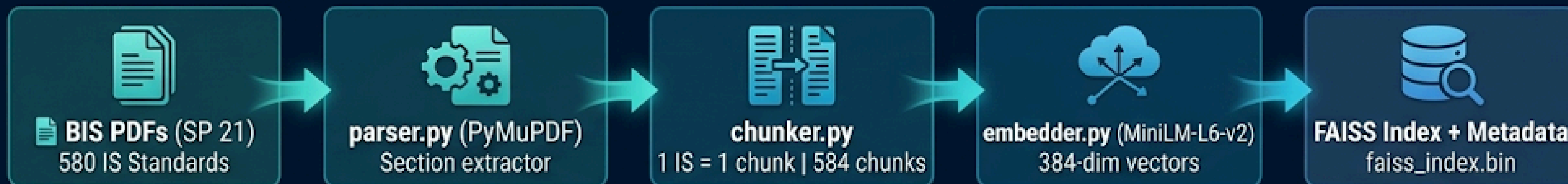
Reproducibility

Portability



03 | System Architecture

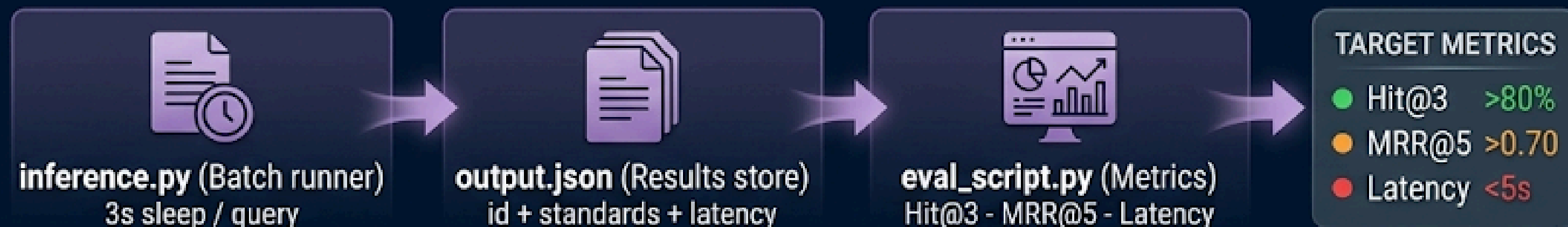
PHASE 1 — OFFLINE INGESTION



PHASE 2 — ONLINE INFERENCE (Per Query)



EVALUATION



Chunking & Retrieval Strategy

584 Chunks · 580 IS Codes

4-Stage Ingestion Pipeline



580 IS Standards

584 Total Chunks

3 Standards Split

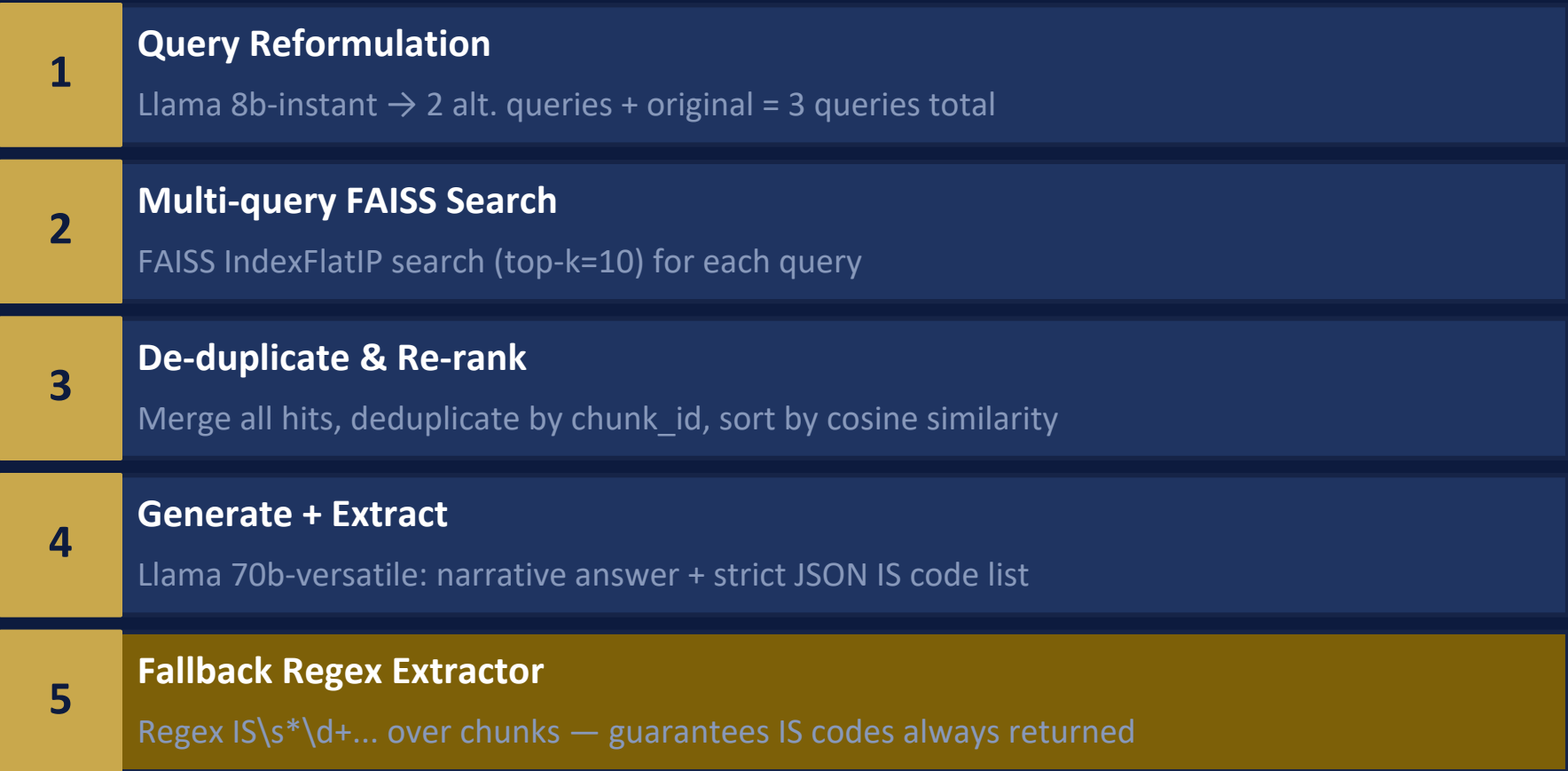
Key Design Decisions

Dual-Field Chunk Design

text_to_embed — Metadata header + content → fed to MiniLM encoder
5-line header: IS Number · Title · Section · Source

content_only — Scope + Requirements only → fed to LLM generation
Saves ~180 tokens of LLM context per query (6 chunks × 30 tokens)

Online Retrieval Flow



384-dim Embeddings

IndexFlatIP · Exact Search

~877 KB FAISS Index

05 | Demo Highlights

Illustrative Example — Cement Manufacturer

User Input

Product: OPC 53 Grade Cement

Use: Structural concrete, RCC

Role: Manufacturer, BIS license pending

Reformulated Queries

Q1: "OPC 53 grade cement compressive strength IS specifications"

Q2: "Portland cement BIS license CRS mandatory requirements"

Q3: "cement fineness setting time soundness test IS 269"

Top Recommendations Returned:

IS 269:2015	IS 1489 Pt.1	IS 4031
Ordinary Portland Cement Mandatory: Yes (CRS) Confidence: High → Apply via BIS Simplified Procedure	Portland Pozzolana Cement Mandatory: Yes (CRS) Confidence: Medium → Apply via BIS Simplified Procedure	Methods of Physical Tests Mandatory: Voluntary Confidence: High → Apply via BIS Simplified Procedure

* Placeholder output for illustrative purposes

Evaluation Results

BIS Hackathon — Public Dataset

10

Queries Evaluated



Hit Rate @3

90%

Target >80%

✓ EXCEEDED

9 of 10 queries returned
a correct IS code in top-3



MRR @5

0.90

Target >0.70

✓ EXCEEDED

Mean Reciprocal Rank
across top-5 retrieved results



Avg Latency

1.39s

Target <5 sec

✓ EXCEEDED

End-to-end wall-clock time
per query (incl. LLM calls)

07 | Impact on MSEs

From weeks of confusion to minutes of clarity.

BEFORE BISfit

- ❌ 6–8 weeks searching BIS website manually
- ❌ Consultant fees of ₹50,000–₹2,00,000
- ❌ Risk of applying wrong / outdated standard
- ❌ No clarity on mandatory vs. voluntary
- ❌ No actionable next step for certification



AFTER BISfit

- ✅ Standard recommendations within few seconds
- ✅ Zero consultant dependency for initial scoping
- ✅ Grounded strictly in official BIS documents
- ✅ Mandatory / CRS flag shown per recommendation
- ✅ Direct certification pathway mapped per standard

BISfit bridges the compliance knowledge gap for 63 million MSEs — making BIS standards accessible, actionable, and fast.

08 | Team & Acknowledgements

BISfit

BIS Standards Recommendation Engine



Shreyash Golhani

Applied AI and Core ML



Vaageesh Kumar Singh

Agentic AI and Product Mindset

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